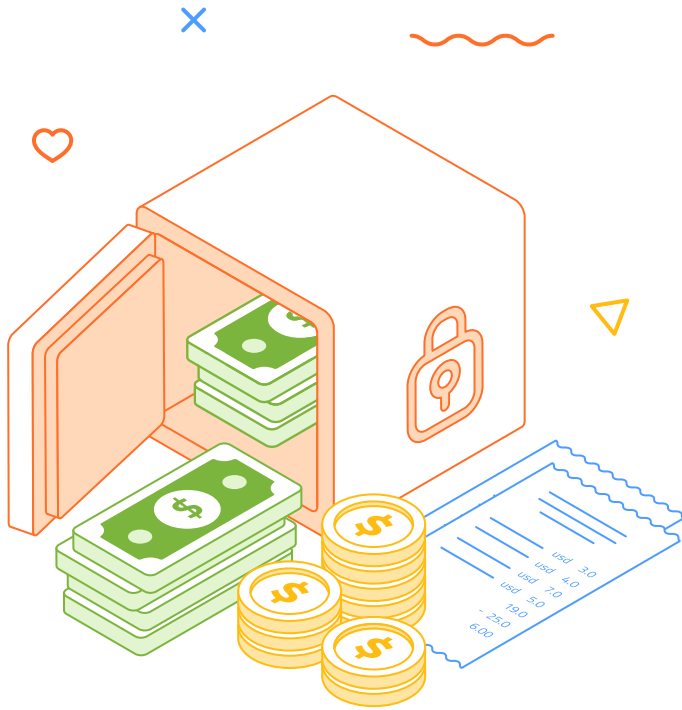




Fleet Procurement Risk Management in the Aviation Industry Using Financial Mathematics

Pavle Radojevic - June 2024

Content



01.

Introduction

Risk Management in the Aviation Industry

02.

Financial Mathematics

Application in the Aviation Industry

03.

Fleet Procurement

04.

Risks in Fleet Procurement



In Fleet Procurement Risk Management

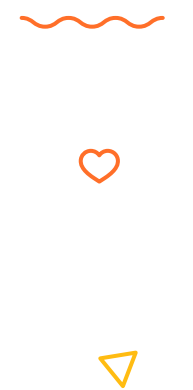
05.

Application of Financial Mathematics

In Fleet Procurement Risk Management

06.

Case Study



Example, Analysis, and Conclusion

Introduction

Risk Management in the
Aviation Industry



01

What is risk management and why is it important in the aviation industry?

- Risk management is the process of identifying, assessing, and controlling risks in order to reduce or minimize the impact of negative events.
- In the aviation industry, risk management refers to all activities carried out to identify, assess, and control risks that may affect flight and passenger safety.
- Risk management in the aviation industry is important because airlines face a wide range of risks that can jeopardize flight and passenger safety.
- These risks include technical problems with aircraft, adverse weather conditions, operational issues, human error, terrorist threats, and others.
- Without adequate risk management, airlines face a significant risk of catastrophic events that can have serious consequences for human lives and property.



Types of Risks in the Aviation Industry (Operational, Financial, Market, etc.)

1. Operational Risk

Human factors (e.g., human error)
Technical problems (e.g., aircraft malfunctions)
Natural disasters (e.g., volcanic eruptions)

2. Financial Risk

Fuel price instability
Currency exchange rate fluctuations
Economic crises

3. Market Risk

Changes in market conditions
Increased competition
Changes in travel trends

4. Reputational Risk

The possibility that poor business practices or negative events can lead to reputational damage, affecting the company's ability to attract new clients and retain existing ones.

5. Security Risk

Terrorist attacks
Sabotage
Natural disasters
Criminal activities



Approaches to Risk Management in the Aviation Industry (Passive, Active, Dynamic)

Passive Risk Management Approach

- Reactive measures taken after an event or incident
- Accident investigation, cause analysis, and prescribing new standards and procedures to prevent similar accidents in the future



Active Risk Management Approach

- Preventive measures taken to prevent risks before they occur
- Regular aircraft inspections, quality control, and crew training
- Often used in combination with the passive approach to ensure maximum safety

Dynamic Risk Management Approach

- Continuous monitoring and risk assessment throughout the aircraft's lifecycle, from design to delivery and maintenance
- Assessment of potential risks and implementation of measures to prevent or minimize risks during the aircraft's lifecycle



02. Financial Mathematics

Application in the Aviation Industry



02

Fundamentals of Financial Mathematics

(Time Value of Money, Discount Rate, etc.)

1. Time Value of Money

Represents the idea that money has greater value now than in the future. The reason is that money can be invested and earn interest, so the amount of money grows over time. When considering financial transactions, the time factor must be taken into account, along with the value of money in different time periods.

3. Interest Rate

Represents the rate at which the value of money increases over a certain time period. Interest rates are used in various financial transactions, such as savings accounts, mortgages, and investments.

2. Discount Rate

Represents the rate at which future values of money are reduced to their present value. This is usually done using the formula for discounting cash flows. This formula takes into account the amount of money, the time at which the money is received, and the discount rate to calculate the present value of money.

4. Future Value of Money

Represents the amount that a sum of money will reach in the future, if invested at a certain interest rate. The future value of money can be calculated using the interest formula based on the investment amount, interest rate, and time period.

Basic Formulas of Financial Mathematics

Discount Rate

$$D = \left(\frac{FV}{PV} \right)^{\left(\frac{1}{n} \right)} - 1$$

FV = Future Value

PV = Present Value

n = Number of periods



Future Value of Money

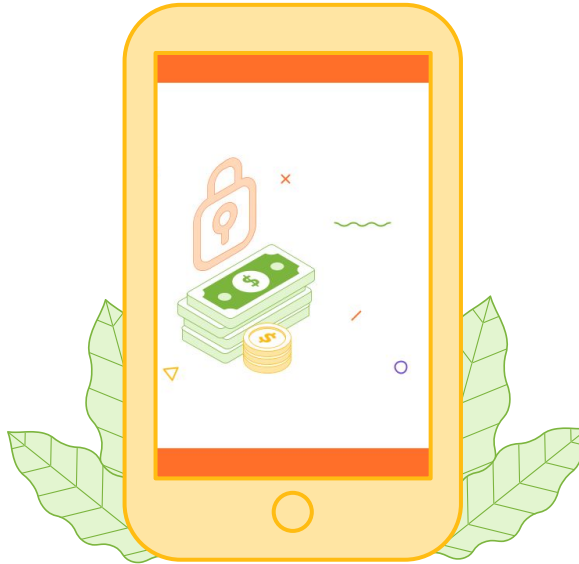
$$FV = PV(1 + rn)$$

FV = Future Value

PV = Present Value
(amount we invest today to earn interest)

r = Simple interest rate

n = Number of periods



Application of Financial Mathematics in the Aviation Industry (Calculating Costs, Revenue, and Profit)

1. Ticket Prices

This includes factors such as fuel costs, aircraft maintenance, staff, and other flight-related costs. Based on these costs, airlines determine the optimal ticket price that will allow them to cover costs and make a profit.



3. Market Analysis

Analysis of the market and identification of best practices for managing flight prices and capacity. For example, if there is high demand for flights in a certain season, airlines can increase ticket prices to maximize revenue.

2. Financial Analysis of Investments

Airlines invest significant amounts of money in purchasing new aircraft and infrastructure. Financial mathematics is used to calculate the potential return on investment and to assess the risks associated with investments.



4. Financial Planning

Planning airline operations and long-term financial projections. This includes budgeting for operating costs, investments, and revenue projections, as well as assessing the risks associated with these plans.

03. Fleet Procurement in the Aviation Industry



03

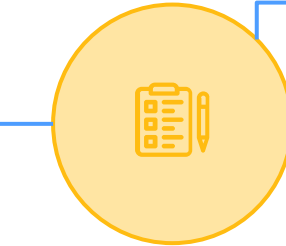
Fleet Procurement Process (Market Research, Aircraft Selection, Negotiation, etc.)

Market Research



Analysis of the current market situation, competition, demand for services, growth forecasts, and trends

Defining Requirements



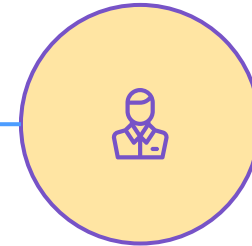
Deciding on the size and type of aircraft, as well as performance, passenger comfort, safety, and price

Aircraft Selection



Reviewing aircraft catalogs, consulting with experts, visiting aircraft manufacturers, and testing aircraft

Negotiation



Negotiating with aircraft manufacturers regarding prices, delivery terms, warranties, etc.

Purchase



Fleet Procurement Costs (Purchase Price, Maintenance Costs, Fuel Costs, etc.)



Purchase Price

This is the price the airline pays for the aircraft. The purchase price includes basic equipment and systems, as well as any additional options and customizations the company chooses. The price varies depending on the type of aircraft, size, and manufacturer.



Fuel Costs

Aircraft use large amounts of fuel, and fuel costs are typically one of the largest costs in the aviation industry. Fuel prices are variable and depend on many factors, including the price of crude oil and political factors in the countries from which the fuel originates.



Maintenance Costs

This includes regular inspections, repairs, servicing, and replacement of worn parts. Maintenance costs depend on the age of the aircraft, number of flight hours, as well as the parts and services used.

Fleet Procurement Financing (Purchase, Leasing, Wet Lease)

Purchase

The company can directly purchase the aircraft and pay for it in full or in installments. This approach allows the company to immediately become the owner of the aircraft, but the cost is higher and there is a risk of the aircraft losing value over time.

Leasing

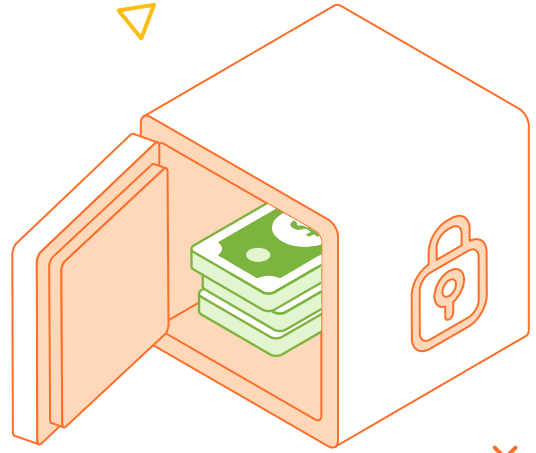
Leasing is a long-term rental of an aircraft. The aircraft is purchased from a leasing company that finances it, while the company renting the aircraft pays monthly installments. After the lease contract expires, the company can decide to return the aircraft or purchase it at the determined market price. This approach reduces the initial costs of acquiring the aircraft and allows the company flexibility in fleet maintenance.

Wet Lease

A "wet lease" represents renting an aircraft together with crew and maintenance. The company renting the aircraft does not have to worry about aircraft maintenance, but only about the operational costs of flying. This approach is most suitable for short-term needs and periodic requirements in the aviation industry (e.g., Air Serbia).



Risks in Fleet Procurement



04

Identification and Assessment of Risks in

× Fleet Procurement in the Aviation Industry

Operational Costs

Procuring a new fleet of aircraft can lead to increased operational costs, such as maintenance, fuel, and staff costs.

Regulatory Changes

Governments and regulatory bodies may introduce new regulations that can affect the procurement and use of aircraft. Therefore, it is important to carefully monitor regulatory changes in order to respond in a timely manner and comply with requirements.



Safety

Procuring a new fleet of aircraft can affect flight safety. Therefore, it is important to carefully assess the quality and safety of the aircraft, as well as the training and certification of personnel who will operate the aircraft.

Long-term Planning

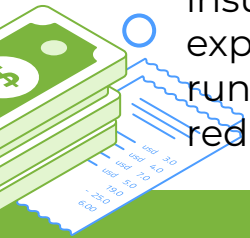
It is important to carefully assess future demand for flights, as well as various factors that may affect the market, such as the economic situation, geopolitical events, and natural disasters.

Risk Mitigation Measures in Fleet Procurement (Insurance, Hedging, etc.)



Insurance

♥ Insurance should cover the damage that the airline may suffer in the event of an accident, damage, or loss of aircraft. Additionally, insurance should cover losses in the event of flight delays or cancellations. Purchasing insurance can be expensive, but in the long run, it is a cheaper way to reduce your risks.



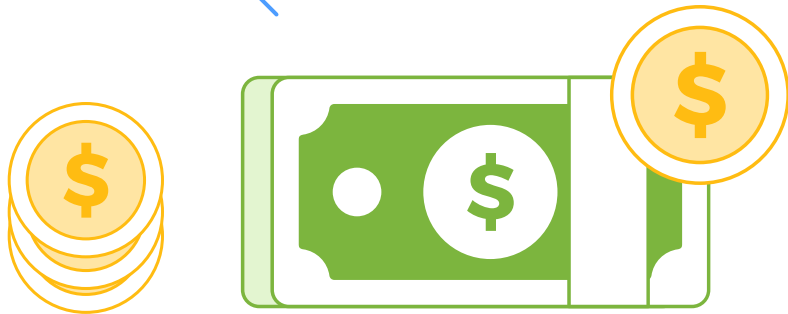
Hedging

✓ Hedging is the process of protection against changes in prices on the fuel, currency, and other markets that may affect the airline's financial position. This is typically achieved using financial instruments such as futures contracts or options.

Market Analysis

This includes monitoring market trends, competition, and demand for flights. By taking these factors into account, the airline can plan its fleet procurement in a way that will allow it to use its resources more efficiently and minimize risks. Also, market analysis can help identify new business opportunities and company expansion.

Application of Financial Mathematics in Fleet Procurement Risk Management



05

Using Financial Mathematics for Risk Management in Fleet Procurement (Option Valuation, Futures Contracts, etc.)

01 Option Valuation

Options are contracts that give the buyer the right, but not the obligation, to buy or sell certain shares or assets at a predetermined price. In the aviation industry, options can be used for fleet purchases, to secure a fixed price for purchasing aircraft in the future.

02 Futures Contracts

Futures contracts are contracts that obligate the buyer to purchase a certain quantity of goods in the future at a predetermined price. In the aviation industry, futures contracts can be used to purchase fuel at a predetermined price, to avoid fuel price fluctuations in the market.

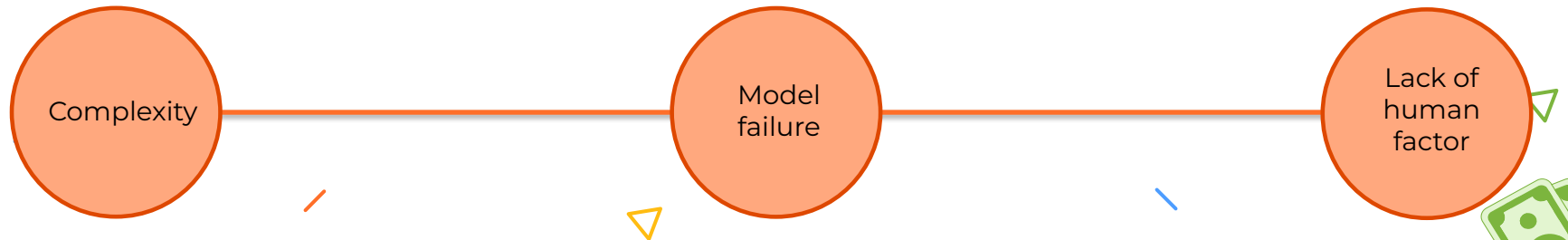
03 Probability Analysis

Probability analysis can be used to assess the likelihood of various events, such as fuel price changes, inflation, or economic instability. In the aviation industry, this analysis can be used to assess risks in fleet procurement and select risk management strategies.

04 Portfolio Optimization

Portfolio optimization can be used to create optimal combinations of different assets in a portfolio, in order to minimize risk and maximize profit. In the aviation industry, this technique can be used to select the optimal combination of aircraft, considering various factors such as fleet size, operational costs, and aircraft performance.

Advantages and Disadvantages of Using Financial Mathematics in Fleet Procurement Risk Management



Case Study



06

Example and Analysis

- 1) Fleet of over 270 aircraft and global passenger transport to over 140 destinations
- 2) Hedging currency risks and financial derivatives - instruments that allow you to trade a specific type of product without becoming the owner, but earning when the product price rises or falls depending on your position.
- 3) Futures contracts for fuel purchases in the future to protect against changes in oil prices on the global market.
- 4) Advanced data analysis methods to predict future trends in the aviation industry and reduce risk in fleet procurement.
- 5) Data analytics to assess when is the best time to procure new aircraft, considering factors such as projected travel growth and economic conditions.
- 6) Fleet diversification. Emirates possesses different types of aircraft, including Airbus A380 and Boeing 777, which allows them to adapt to market changes.



Boeing 777-300ER



Airbus A380-800



Discussion of Case Study Results and Conclusions on the Application of Financial Mathematics in Fleet Procurement Risk Management in the Aviation Industry

- The Emirates case study shows that the application of financial mathematics in risk management in fleet procurement in the aviation industry can be an effective approach. Emirates has implemented various risk management strategies, including currency risk hedging, fleet diversification, and data analysis, which has led to risk reduction and optimization of investment returns.
- Using financial instruments, Emirates has successfully protected itself against changes in oil prices on the global market and reduced currency risks. Furthermore, fleet diversification allows Emirates to adapt to market changes and reduce the risk of economic changes that may affect certain types of aircraft. Data analysis enables Emirates to predict future trends in the aviation industry and reduce risk in the procurement of new aircraft.
- As a result of implementing these strategies, Emirates has achieved more efficient risk management in fleet procurement and improved investment returns. This shows that financial mathematics can be useful in risk management in fleet procurement in the aviation industry.

**Thank you for
your attention!**

